

SKILL ASSESSMENT - 2

19ME302 - ENGINEERING DRAWING

PROJECT REPORT

WOODEN OUDH BURNER STAND

Design Study and Engineering Representation

Submitted by

MOHAMED SHABEEH M

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ABSTRACT

This report presents the design study of a wooden oudh burner stand, a pedestal-style product intended to support a metal receptacle for aromatic incense. The supplied photographs show a circular wooden upper body, a shaped central stem, a broad circular base and a metal bowl seated in the top opening. Decorative circumferential grooves and the natural appearance of wood contribute to the product's visual character.

The project connects a practical wooden product with engineering drawing concepts, including symmetry, orthographic views, sectional representation, dimensioning and component identification. A proposed dimension schedule and a suitable fabrication sequence are provided for academic design development. The wooden structure serves as the support; it must not be treated as the combustion chamber.

The report distinguishes visible features from proposed specifications. Exact measurements, wood species, joining details, manufacturing records and test results were not supplied. Accordingly, dimensions and fabrication details are presented as proposals, while evaluation is limited to photographic observations. Thermal separation between the metal bowl and wood requires verification before any heated use.

Keywords

Wooden oudh burner stand; incense holder; engineering drawing; wood turning; pedestal; metal insert.

CONTENTS

Section	Page
1. Introduction, aim and objectives	3
2. Product description and materials	4
3. Proposed dimensions and drawing views	5
4. Suggested fabrication and working principle	6
5. Product photographs	7
6. Evaluation, maintenance and improvements	8
7. Conclusion and reference material	9

1. INTRODUCTION, AIM AND OBJECTIVES

1.1 Introduction

An oudh burner stand is a support used with a receptacle for burning aromatic material. For this project, the suitable product name is “Wooden Oudh Burner Stand”. This identifies the main structural material and intended function without suggesting that incense should be burned directly on the wood.

The photographed product has a roughly rotationally symmetric form. Its upper bowl-shaped housing is connected to a turned-looking central stem and a flared base. This makes the product appropriate for studying centre lines, circular features, curved profiles and the relationship between front and top views.

1.2 Aim

To study and represent a wooden oudh burner stand through engineering drawings, identify its major components and describe a suitable method of fabrication and evaluation.

1.3 Objectives

- Identify the upper housing, stem, base and metal bowl in the supplied product photographs.
- Prepare schematic front and top views and explain the side and sectional views.
- Develop a clearly labelled proposed dimension schedule for a similar model.
- Describe material selection, shaping, finishing and assembly operations.
- Recognise the importance of stability, bowl retention and thermal separation.

1.4 Scope

This is a photograph-based product study for Engineering Drawing. It includes a suggested construction method and inspection plan. It does not establish the actual product dimensions, load capacity, temperature limits, manufacturing cost or suitability for live burning.

2. PRODUCT DESCRIPTION AND MATERIALS

2.1 Construction features

Part	Observed form and function
Upper wooden housing	Rounded circular body with external grooves; supports and surrounds the metal bowl.
Central stem	Narrow shaped support between the upper housing and base; transfers the supported weight downward.
Circular base	Flared lower portion providing a footprint on the supporting surface.
Metal bowl / insert	Open metal receptacle seated in the upper opening; dark residue is visible inside.
External surface	Brown wood appearance with grooves, visible marks and areas of finish wear.

2.2 Suggested material selection

For a similar model, a sound, seasoned solid-wood blank is suitable for the supporting body. Select material without loose knots, splits or decayed areas. The species of the photographed wood cannot be identified reliably from the images. A compatible metal receptacle must be selected for the intended service; its alloy and thickness are also unconfirmed.

The visible metal bowl is not proof of adequate heat protection. Metal conducts heat to adjacent materials. A designed non-combustible insulating arrangement and appropriate separation from wood would need to be specified and checked before heated operation. Such insulation is not identifiable in the supplied photographs.

2.3 Tools for a proposed model

Steel rule, vernier caliper, pencil, marking gauge, saw, wood lathe with suitable tooling, holding fixtures, drilling tools where required, sandpaper and workshop protective equipment. Tool selection depends on whether the body is made as one piece or assembled from separate parts.

3. PROPOSED DIMENSIONS AND DRAWING VIEWS

The following values are illustrative design dimensions for a similar model, not measurements of the photographed product. All dimensions are in millimetres. Measure the actual article before preparing a production drawing.

Feature	Proposed value
Overall height / upper housing height	250 / 75
Upper housing outside diameter	150
Stem height / minimum diameter	125 / 35
Base height / maximum diameter	50 / 125
Bowl seating opening / recess depth	120 / 50; final fit to selected insert

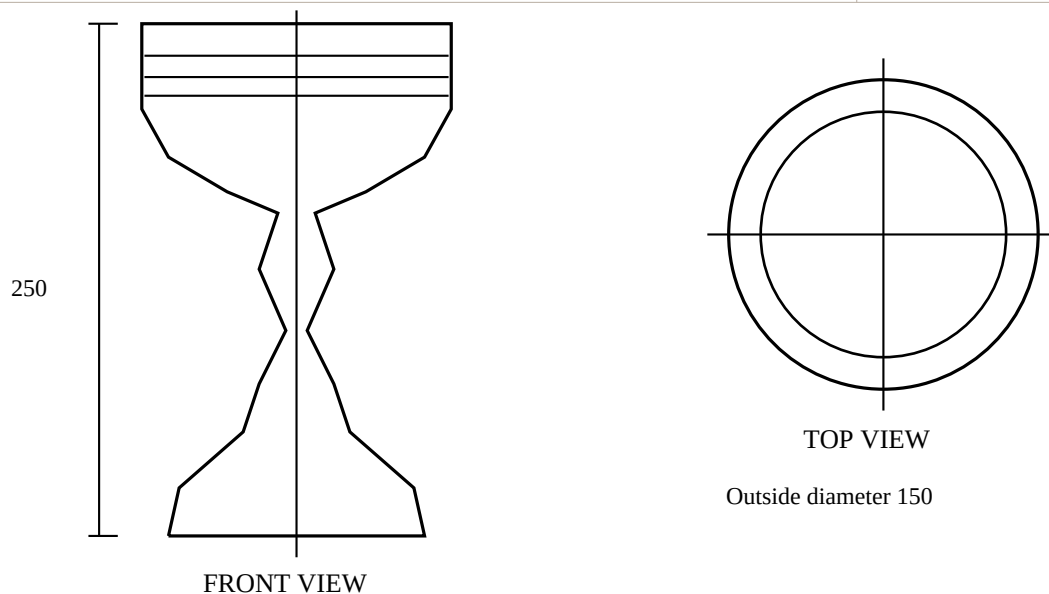


Figure 1. Schematic orthographic views, not to scale. Decorative profiles are indicative; the drawing does not show verified assembly joints.

Side view: nominally matches the front outline for an axisymmetric model. **Sectional view:** should show the actual recess, bowl seat, remaining wood thickness and verified thermal separation. **Isometric view:** should show the circular openings as ellipses and the complete pedestal form.

4. SUGGESTED FABRICATION AND WORKING PRINCIPLE

4.1 Proposed fabrication procedure

1. Measure the reference product or approve a dimensioned concept. Select the metal insert and design its seating and thermal separation before cutting the wood.
2. Select sound, seasoned wood and mark the centre line, overall length and maximum diameter. Allow suitable machining allowance.
3. Cut the blank and secure it using an appropriate workshop holding method. Check the setup under supervision.
4. Shape the broad base, central stem and upper body. A wood lathe is a suitable proposed process for the circular profile; the original process is not documented.
5. Form the upper recess to suit the insert and the specified insulating arrangement. Retain sufficient wood around and beneath the recess.
6. Cut shallow decorative grooves without excessively reducing the stem or housing thickness. Smooth the edges and surfaces.
7. If separate components are used, design and make aligned joints with suitable retention. The photographs do not establish whether the original body is one piece or assembled.
8. Check the base for flatness and the insert for secure seating. Finish the exterior with a suitable coating, keeping it away from the hot zone.
9. Inspect dimensions, joints, finish and stability. Complete thermal assessment before considering the model ready for heated service.

4.2 Working principle

The upper housing supports the insert, the stem carries its weight and the base distributes that weight to the supporting surface. For stability, the combined centre of gravity must remain over the base footprint. A broad base helps, but does not itself prove resistance to tipping.

In intended use, the aromatic material belongs in a suitable heat-containing receptacle. Heat can pass from the bowl towards the wooden support, so thermal separation is a design requirement. No electrical heating system is visible in the supplied product.

5. PRODUCT PHOTOGRAPHS



Figures 2 and 3. Product photographs with edited neutral backgrounds, showing the wooden pedestal body and metal insert from different viewpoints.

The overall view records the grooved upper body, shaped stem and flared base. The overhead view records the circular bowl, surrounding wooden rim and dark residue inside the receptacle.

These photographs are the primary evidence for the product description. They do not contain a measuring scale and do not reveal concealed joints or any insulating layer beneath the bowl. The backgrounds have been edited for presentation; the original photographs remain the basis for the product observations.

6. EVALUATION, MAINTENANCE AND IMPROVEMENTS

6.1 Observation and verification plan

Check	Evidence / required verification
Appearance	Pedestal form and decorative grooves are visible. Surface wear and marks are present.
Dimensions	Not measured. Check height, diameters and recess with measuring tools.
Stability	Not tested. Assess rocking and tipping tendency on a rigid, level surface with the intended insert.
Bowl retention	Insert is visible in position. Seating clearance and resistance to displacement require inspection.
Thermal performance	Not tested. Check heat transfer, insulation and the risk of wood degradation before heated service.
Durability	Not established. Inspect for cracks, loose joints and changes after repeated use.

6.2 Care and use considerations

- Keep the wooden body dry and inspect it for splitting, loosening or charring.
- Clean the receptacle only after it has cooled completely; remove residue without damaging its seat.
- Do not place burning material directly against wood or use the product on fabric.
- Any heated use requires a stable, non-combustible surface, adequate ventilation and continuous supervision. Do not carry it while hot.

6.3 Advantages, limitations and future work

The simple circular form provides a useful exercise in turning, profile drawing and component representation. The wooden exterior offers decorative value. Limitations include a relatively narrow stem, unverified bowl retention and unverified heat isolation. Future development should prioritise a properly specified insulated receptacle, a stable base and measured inspection results.

No bills or material quantities were supplied. A defensible cost estimate should separately record wood, insert, insulation, finishing consumables and machining labour; a manufacturing total is therefore not claimed.

7. CONCLUSION AND REFERENCE MATERIAL

7.1 Result and conclusion

The wooden outh burner stand has been documented as an Engineering Drawing product study using the supplied photographs. Its principal visible features are a circular upper wooden housing, shaped central stem, flared base and metal receptacle. The report provides a component description, illustrative dimensions, schematic drawing views and a proposed fabrication sequence.

The project demonstrates how a practical product can be analysed through geometry, material selection, load transfer and engineering representation. It also shows why the support structure and the heat-containing component must be considered separately during design.

The available evidence supports a description of the product's appearance and intended arrangement. It does not establish tested performance. Actual dimensions, bowl fit, structural stability and thermal protection should be verified before a final manufacturing drawing or operational approval is issued.

7.2 Relevant technical literature

The USDA Forest Products Laboratory's Wood Handbook provides engineering background on wood properties, moisture control, finishing and fire behaviour. These topics are relevant to selecting the wooden body, maintaining the bowl fit and assessing heat exposure. Its guidance is general; it does not validate this particular burner design. [1]

The American Association of Woodturners provides practical guidance for lathe work. For the proposed fabrication process, this includes secure mounting, a full face shield, suitable starting speed and stopping the machine before adjusting the tool rest. These recommendations apply to shaping the pedestal and upper housing. [2]

7.3 References

[1] Forest Products Laboratory. (2021). *Wood Handbook: Wood as an Engineering Material*. USDA Forest Service. Relevant chapters: 4, Moisture Relations and Physical Properties of Wood; 13, Drying and Control of Moisture Content and Dimensional Changes; 16, Finishing Wood; and 18, Fire Safety of Wood Construction. [Official publication and chapter downloads](#).

[2] American Association of Woodturners. *Woodturning Safety*, including AAW Safety Recommendations and the *Safety Guidebook for Woodturners*. [Official safety guidance and guidebook access](#). Accessed 14 September 2026.